1 Indeterminate Forms

Suppose we would like to know the value of the limit

$$\lim_{x\to 1}\frac{\ln(x)}{x-1}, \tag{1}$$

if it exists. We cannot apply the Quotient Limit Law, since $\lim_{x\to 1}(x-1)=0$. Since $\lim_{x\to 1}\ln(x)=0$ as well, the limit in Equation (1) is an indeterminate form of type $\frac{0}{0}$. We have encountered limits of this type before. For rational functions, we can cancel common factors:

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=\lim_{x\to 1}\frac{(x+1)(x-1)}{x-1}=\lim_{x\to 1}(x+1)=2.$$

We used the Squeeze Theorem to show that

$$\lim_{x\to 0}\frac{\sin(x)}{x}=1.$$

Neither of these methods will work for limits such as the one in Equation (1). We now introduce a systematic method, known as *L'Hôpital's Rule*, for the evaluation of indeterminate forms.

Another situation in which a limit is not obvious occurs when we look for a horizontal asymptote, such as in

$$\lim_{x\to\infty}\frac{\ln(x)}{x-1}. \tag{2}$$

Since both $\lim_{x\to\infty}\ln(x)=\infty$ and $\lim_{x\to\infty}(x-1)=\infty$, this limit is an indeterminate form of type $\frac{\infty}{\infty}$. We have also encountered such limits before. For rational functions, we can find the limit by dividing the numerator and denominator by the highest power of x that occurs in the denominator. For example,

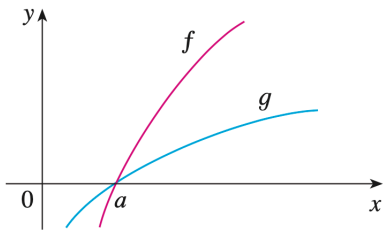
$$\lim_{x\to\infty}\frac{x^2-1}{2x^2+1}=\lim_{x\to\infty}\frac{1-\frac{1}{x^2}}{2+\frac{1}{x^2}}=\frac{\lim_{x\to\infty}(1-\frac{1}{x^2})}{\lim_{x\to\infty}(2+\frac{1}{x^2})}=\frac{1-0}{2+0}=\frac{1}{2}.$$

This method does not work for limits such as the one in Equation (2), but L'Hôpital's Rule also applies to this type of intederminate form.

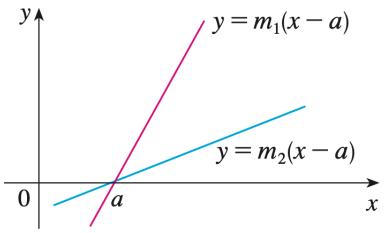
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2 Motivating L'Hôpital's Rule

Consider the limit $\lim_{x\to a}\frac{f(x)}{g(x)}$ for the two functions whose graphs are shown below.



Since $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, the limit $\lim_{x\to a}\frac{f(x)}{g(x)}$ is an indeterminate form of type $\frac{0}{0}$. If both f and g are differentiable at a, then near a they are both well approximated by their tangent lines at a, which are shown below. If the functions actually were linear, as in the second graph,



then we would have

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{m_1(x-a)}{m_2(x-a)} = \lim_{x\to a}\frac{m_1}{m_2} = \lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

3 L'Hôpital's Rule

Theorem 3.1 (L'Hôpital's Rule)**.** *Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a itself). Suppose that*

$$\lim_{x\to a} f(x) = 0 \quad \textit{and} \quad \lim_{x\to a} g(x) = 0$$

or that

$$\lim_{x\to a} f(x) = \pm\infty \quad \textit{and} \quad \lim_{x\to a} g(x) = \pm\infty.$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is $\pm\infty$).

L'Hôpital's Rule is also true for one-sided limits and for limits at $\pm\infty$; that is, "$x \to a$" can be replaced with any of the symbols "$x \to a^+, x \to a^-, x \to \infty$, or $x \to -\infty$".

Before proving L'Hôpital's Rule, we work several examples.

Example 3.2. Find $\lim_{x\to 1} \frac{\ln(x)}{x-1}$.

Solution. Since

$$\lim_{x\to 1} \ln(x) = \ln(1) = 0 \quad \text{and} \quad \lim_{x\to 1} (x-1) = 1-1 = 0,$$

we can apply L'Hôpital's Rule:

$$\lim_{x\to 1} \frac{\ln(x)}{x-1} = \lim_{x\to 1} \frac{\frac{d}{dx}\ln(x)}{\frac{d}{dx}(x-1)} = \lim_{x\to 1} \frac{1/x}{1} = \lim_{x\to 1} \frac{1}{x} = 1.$$

1. Calculate $\lim_{x\to\infty} \frac{e^x}{x^2}$.

   Indeterminate of type $\frac{\infty}{\infty}$, so $\lim_{x\to\infty} \frac{e^x}{x^2} = \lim_{x\to\infty} \frac{e^x}{2x} = \lim_{x\to\infty} \frac{e^x}{2} = \infty$.

2. Calculate $\lim_{x\to\infty} \frac{\ln(x)}{\sqrt[3]{x}}$.

   Indeterminate of type $\frac{\infty}{\infty}$, so $\lim_{x\to\infty} \frac{\ln(x)}{\sqrt[3]{x}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{3}x^{-2/3}} = \lim_{x\to\infty} \frac{3x^{2/3}}{x} = \lim_{x\to\infty} \frac{3}{x^{1/3}} = 0.$

3. Calculate $\lim_{x\to 0}\dfrac{\tan(x)-x}{x^3}$.
Indeterminate of type $\frac{0}{0}$, so

SOLUTION Noting that both $\tan x - x \to 0$ and $x^3 \to 0$ as $x \to 0$, we use l'Hospital's Rule:

$$\lim_{x\to 0}\frac{\tan x - x}{x^3} = \lim_{x\to 0}\frac{\sec^2 x - 1}{3x^2}$$

Since the limit on the right side is still indeterminate of type $\frac{0}{0}$, we apply l'Hospital's Rule again:

$$\lim_{x\to 0}\frac{\sec^2 x - 1}{3x^2} = \lim_{x\to 0}\frac{2\sec^2 x \tan x}{6x}$$

Because $\lim_{x\to 0}\sec^2 x = 1$, we simplify the calculation by writing

$$\lim_{x\to 0}\frac{2\sec^2 x \tan x}{6x} = \frac{1}{3}\lim_{x\to 0}\sec^2 x \cdot \lim_{x\to 0}\frac{\tan x}{x} = \frac{1}{3}\lim_{x\to 0}\frac{\tan x}{x}$$

We can evaluate this last limit either by using l'Hospital's Rule a third time or by writing $\tan x$ as $(\sin x)/(\cos x)$ and making use of our knowledge of trigonometric limits. Putting together all the steps, we get

$$\lim_{x\to 0}\frac{\tan x - x}{x^3} = \lim_{x\to 0}\frac{\sec^2 x - 1}{3x^2} = \lim_{x\to 0}\frac{2\sec^2 x \tan x}{6x}$$

$$= \frac{1}{3}\lim_{x\to 0}\frac{\tan x}{x} = \frac{1}{3}\lim_{x\to 0}\frac{\sec^2 x}{1} = \frac{1}{3}$$

4. Note that L'Hôpital's Rule only applies to certain indeterminate limits, as $\lim_{x\to a}\frac{f(x)}{g(x)} \neq \lim_{x\to a}\frac{f'(x)}{g'(x)}$ in general. For instance, show that

$$\lim_{x\to 2}\frac{x+1}{x} \neq \lim_{x\to 2}\frac{\frac{d}{dx}(x+1)}{\frac{d}{dx}x}.$$

$$\lim_{x\to 2}\frac{x+1}{x} = \frac{2+1}{2} = \frac{3}{2},$$

while

$$\lim_{x\to 2}\frac{\frac{d}{dx}(x+1)}{\frac{d}{dx}x} = \lim_{x\to 2}\frac{1}{1} = \lim_{x\to 2} 1 = 1 \neq \lim_{x\to 2}\frac{x+1}{x}.$$

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4 Indeterminate Products

Example 4.1. Evaluate $\lim_{x\to 0^+} x\ln(x)$.

Solution. Since $\lim_{x\to 0^+} x = 0$ and $\lim_{x\to 0^+} \ln(x) = -\infty$, this is an indeterminate form of type $0\cdot\infty$. To solve, write $x = \frac{1}{1/x}$ to convert it to an indeterminate form of type $\frac{\infty}{\infty}$ and use L'Hôpital's Rule:

$$\lim_{x\to 0^+} x\ln(x) = \lim_{x\to 0^+} \frac{\ln(x)}{1/x} = \lim_{x\to 0^+} \frac{\frac{d}{dx}\ln(x)}{\frac{d}{dx}1/x} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = \lim_{x\to 0^+} (-x) = 0.$$

Note that we could have also written $\ln(x) = \frac{1}{\ln(x)}$ to convert the limit above to an indeterminate form of type $\frac{0}{0}$, but then applying L'Hôpital's Rule would have resulted in a more complicated expression.

4. Calculate $\lim_{x\to 0} x\cot(x)$.

   This is indeterminate of type $0\cdot\infty$, so $\lim_{x\to 0} \frac{\cot(x)}{1/x}$ is indeterminate of type $\frac{\infty}{\infty}$. It follows that

$$\lim_{x\to 0} x\cot(x) = \lim_{x\to 0} \frac{\cot(x)}{1/x} = \lim_{x\to 0} \frac{-\csc^2(x)}{-\frac{1}{x^2}} = \lim_{x\to 0} \frac{x^2}{\sin^2(x)} = \lim_{x\to 0} \left(\frac{x}{\sin(x)}\right)^2$$
$$= \left(\lim_{x\to 0} \frac{x}{\sin(x)}\right)^2 = \left(\lim_{x\to 0} \frac{1}{\frac{\sin(x)}{x}}\right)^2 = \left(\frac{1}{\lim_{x\to 0} \frac{\sin(x)}{x}}\right)^2 = 1.$$

5. Calculate $\lim_{x\to\infty} x\sin(\frac{1}{x})$.

   $\lim_{x\to\infty} x\sin(\frac{1}{x})$ is indeterminate of type $0\cdot\infty$, so $\lim_{x\to\infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}}$ is indeterminate of type 0/0. It follows that

$$\lim_{x\to\infty} x\sin(\frac{1}{x}) = \lim_{x\to\infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}} = \lim_{x\to\infty} \frac{\cos(\frac{1}{x})(-\frac{1}{x^2})}{-\frac{1}{x^2}} = \lim_{x\to\infty} \cos(\frac{1}{x}) = \cos(\lim_{x\to\infty} \frac{1}{x}) = \cos(0) = 1.$$

5 Indeterminate Differences

If $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$, then the limit

$$\lim_{x\to a}(f(x) - g(x))$$

is an indeterminate form of type $\infty - \infty$. In this case, we convert the function into a quotient.

Example 5.1. Compute $\lim_{x\to(\pi/2)^-}(\sec(x) - \tan(x))$.

Solution. Since $\lim_{x\to(\pi/2)^-} \sec(x) = \infty$ and $\lim_{x\to(\pi/2)^-} \tan(x) = \infty$, the limit is indeterminate of type $\infty - \infty$. Since $\sec(x) = \frac{1}{\cos(x)}$ and $\tan(x) = \frac{\sin(x)}{\cos(x)}$, we can put these over a common denominator of $\cos(x)$ so that

$$\lim_{x\to(\pi/2)^-}(\sec(x) - \tan(x)) = \lim_{x\to(\pi/2)^-} \frac{1-\sin(x)}{\cos(x)}$$

which is indeterminate of type $\frac{0}{0}$. Applying L'Hôpital's Rule gives

$$\begin{aligned}\lim_{x\to(\pi/2)^-}(\sec(x) - \tan(x)) &= \lim_{x\to(\pi/2)^-} \frac{1-\sin(x)}{\cos(x)} = \lim_{x\to(\pi/2)^-} \frac{\frac{d}{dx}(1-\sin(x))}{\frac{d}{dx}\cos(x)} = \lim_{x\to(\pi/2)^-} \frac{-\cos(x)}{-\sin(x)} \\ &= \lim_{x\to(\pi/2)^-} \cot(x) = 0.\end{aligned}$$

6. Calculate $\lim_{x\to 0^+}\left(\frac{1}{x^2}-\frac{1}{\tan(x)}\right)$.

By combining the fractions, we can write the function as a quotient. Since the least common denominator is $x^2\tan x$, we have

$$\frac{1}{x^2}-\frac{1}{\tan x}=\frac{(\tan x)-x^2}{x^2\tan x}.$$

As $x\to 0^+$, the numerator $\tan x - x^2 \to 0$ and the denominator $x^2\tan x \to 0$. Therefore, we can apply L'Hôpital's rule. Taking the derivatives of the numerator and the denominator, we have

$$\lim_{x\to 0^+}\frac{(\tan x)-x^2}{x^2\tan x}=\lim_{x\to 0^+}\frac{\left(\sec^2 x\right)-2x}{x^2\sec^2 x+2x\tan x}.$$

As $x\to 0^+$, $\left(\sec^2 x\right)-2x\to 1$ and $x^2\sec^2 x+2x\tan x\to 0$. Since the denominator is positive as x approaches zero from the right, we conclude that

$$\lim_{x\to 0^+}\frac{\left(\sec^2 x\right)-2x}{x^2\sec^2 x+2x\tan x}=\infty.$$

Therefore,

$$\lim_{x\to 0^+}\left(\frac{1}{x^2}-\frac{1}{\tan x}\right)=\infty.$$

6 Indeterminate Powers

Several indeterminate forms arise from the limit

$$\lim_{x\to a}[f(x)]^{g(x)}.$$

(i) $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ type 0^0

(ii) $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = 0$ type ∞^0

(iii) $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \pm\infty$ type 1^∞

Each of these three cases can be treated either by taking the natural logarithm:

$$\text{If } y = [f(x)]^{g(x)}, \text{then } \ln(y) = g(x)\ln(f(x))$$

or by writing the function as an exponential:

$$[f(x)]^{g(x)} = e^{g(x)\ln(f(x))}.$$

Either method leads us to an indeterminate product $g(x)\ln(f(x))$, which is of type $0\cdot\infty$. Of course, this only makes sense if $y = [f(x)]^{g(x)} > 0$ on an open interval containing a, except perhaps at a itself.

Example 6.1. Calculate $\lim_{x\to 0^+}(1+\sin(4x))^{\cot(x)}$.

Solution. Since $\lim_{x\to 0^+}(1+\sin(4x)) = 1$ and $\lim_{x\to 0^+}\cot(x) = \infty$ is an indeterminate form of type 1^∞. If we let

$$y = (1+\sin(4x))^{\cot(x)}$$

then

$$\ln(y) = \cot(x)\ln(1+\sin(4x)),$$

so L'Hôpital's Rule gives

$$\lim_{x\to 0^+}\ln(y) = \lim_{x\to 0^+}\cot(x)\ln(1+\sin(4x)) = \lim_{x\to 0^+}\frac{\ln(1+\sin(4x))}{\tan(x)} = \lim_{x\to 0^+}\frac{\frac{4\cos(4x)}{1+\sin(4x)}}{\sec^2(x)} = 4,$$

and therefore

$$\lim_{x\to 0^+}(1+\sin(4x))^{\cot(x)} = \lim_{x\to 0^+} y = \lim_{x\to 0^+} e^{\ln(y)} = e^{\lim_{x\to 0^+}\ln(y)} = e^4.$$

7. Find $\lim_{x\to 0^+} x^{\frac{1}{x}}$.

Let $y = x^{1/x}$. Then,

$$\ln\left(x^{1/x}\right) = \frac{1}{x}\ln x = \frac{\ln x}{x}.$$

We need to evaluate $\lim_{x\to\infty} \frac{\ln x}{x}$. Applying L'Hôpital's rule, we obtain

$$\lim_{x\to\infty} \ln y = \lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0.$$

Therefore, $\lim_{x\to\infty} \ln y = 0$. Since the natural logarithm function is continuous, we conclude that

$$\ln\left(\lim_{x\to\infty} y\right) = 0,$$

which leads to

$$\lim_{x\to\infty} y = \lim_{x\to\infty} \frac{\ln x}{x} = e^0 = 1.$$

Hence,

$$\lim_{x\to\infty} x^{1/x} = 1.$$